

COMPUTER NETWORKS AND SECURITY --- SAMPLE SOLUTION

Paper Code: -38 | Total Marks: 70 | Time: 2½ Hours

[6262]

Q1) a) Circuit Switching vs Packet Switching

[6]

Parameter	Circuit Switching	Packet Switching
Resource allocation	Dedicated path reserved for entire call	Resources shared on demand
Path	Fixed, established before communication	Dynamic, each packet may take different path
Delay	Fixed (propagation only)	Variable (queuing + propagation)
Efficiency	Wasted bandwidth during silence	Efficient bandwidth utilization
Store-and-forward	No	Yes
Call setup	Required (connection establishment)	Not required (connectionless)
Example	Traditional telephone network	Internet (IP)

Q1) b) Routing Information Protocol (RIP)

[6]

RIP is a distance-vector routing protocol used in small to medium-sized networks.

- **Metric:** Hop count (max 15 hops --- 16 = infinity/unreachable)
- **Algorithm:** Bellman-Ford distance vector
- **Updates:** Every 30 seconds (full routing table broadcast)
- **Port:** UDP port 520
- **Versions:** RIPv1 (classful), RIPv2 (classless, supports CIDR/VLSM)

Features:

- **Split horizon:** Never advertise a route back on the interface it was learned from
- **Poison reverse:** Advertise failed routes with metric 16
- **Hold-down timer:** 180 seconds before accepting new routes for a failed network

Limitations: Slow convergence, limited to 15 hops, high bandwidth usage for updates.

Q1) c) IP Subnetting

[6]

Given: 192.168.5.71 / 26

/26 means 26 network bits, 6 host bits.

i) Subnet mask: 255.255.255.192 (binary: 11111111.11111111.11111111.11000000)

ii) Network address:

- 71 in binary = 01000111
- With /26 mask (11000000): 01000000 = 64
- **Network address:** 192.168.5.64

iii) First usable IP: 192.168.5.65

iv) Last usable IP: 192.168.5.126 (broadcast is 192.168.5.127)

v) Broadcast address: 192.168.5.127

[ANSWER BOX]

Subnet mask: 255.255.255.192
 Network address: 192.168.5.64
 First usable: 192.168.5.65
 Last usable: 192.168.5.126
 Broadcast: 192.168.5.127

Q2) a) IPv6 Header Format

[6]

IPv6 Header (40 bytes fixed):

Version (4 bit)	Traffic Class (8 bit)	Flow Label (20 bit)
Payload Length (16 bit)	Next Header (8 bit)	Hop Limit (8 bit)
Source Address (128 bit)		
Destination Address (128 bit)		

Improvements over IPv4:

- Larger address space (128-bit vs 32-bit)
- Simplified header (no checksum, no fragmentation in routers)
- Built-in security (IPSec mandatory)
- Better multicast support (anycast introduced)
- No NAT required (true end-to-end connectivity)
- Autoconfiguration (stateless address autoconfiguration/SLAAC)

Q2) b) Border Gateway Protocol (BGP)

[6]

BGP is the path-vector routing protocol that governs routing between Autonomous Systems (AS) on the internet.

- **Type:** Exterior Gateway Protocol (EGP)
- **Transport:** TCP port 179
- **Metric:** Path attributes (AS-path, next-hop, local preference)
- **Versions:** BGP-4 (current)

Key concepts:

- **eBGP:** Between different ASes
- **iBGP:** Within the same AS
- **Path attributes:** AS_PATH, NEXT_HOP, LOCAL_PREF, MED
- **Route selection:** Shortest AS_PATH → lowest MED → eBGP > iBGP

Advantages: Policy-based routing, loop-free (AS_PATH contains all ASes), scalable for global internet.

Q2) c) Network Layer Functions

[6]

1. **Logical Addressing:** Assigns IP addresses (IPv4/IPv6) to uniquely identify hosts
2. **Routing:** Determines the best path from source to destination
3. **Forwarding:** Moves packets from input to output port of a router
4. **Fragmentation & Reassembly:** Splits packets when MTU differs across networks
5. **Error Handling:** ICMP messages for unreachable hosts, TTL exceeded
6. **Quality of Service (QoS):** Traffic prioritization, congestion notification

Q3) a) TCP Header Format

[6]

Source Port (16 bit)		Destination Port (16 bit)					
Sequence Number (32 bit)							
Acknowledgment Number (32 bit)							
Data Offset 4bit	Reserved (3 bits)	U A	P S	R S	S Y	F I	Window (16 bit)
		R G	S H	S T	Y N	I N	
Checksum (16 bit)				Urgent Pointer (16 bit)			
Options (0-40 bytes)							
Data							

Flags: URG, ACK, PSH, RST, SYN, FIN --- used for connection management.

Q3) b) Transport Layer Services

[6]

1. **Process-to-Process Delivery:** Data delivered to the correct application (via port numbers)
2. **Connection Establishment/Release:** TCP uses 3-way handshake; UDP is connectionless
3. **Flow Control:** Prevents sender from overwhelming receiver (TCP sliding window)
4. **Error Control:** Checksum verification, retransmission of lost segments
5. **Congestion Control:** TCP uses AIMD (Additive Increase Multiplicative Decrease)
6. **Multiplexing/Demultiplexing:** Multiple applications share the same network layer connection

Q3) c) UDP Header Parsing

[6]

UDP Header Format: Source Port(2B) | Dest Port(2B) | Length(2B) | Checksum(2B)

Hex dump: e2 a7 00 0d 00 20 74 9e 0e ff 00 00 00 01 00 00 00

First 8 bytes = UDP header:

- **Source Port:** e2 a7 = 58023 (decimal)
- **Dest Port:** 00 0d = 13 (decimal) --- Daytime protocol
- **Length:** 00 20 = 32 bytes (total UDP datagram including header)
- **Checksum:** 74 9e --- (validated at receiver)

[ANSWER BOX]

Source Port: 58023
Destination Port: 13 (Daytime)
Total Length: 32 bytes
Data Length: 24 bytes (32 - 8 header)

Q4) a) UDP Header Format

[6]

Source Port (16 bit)		Destination Port (16 bit)					
Length (16 bit)		Checksum (16 bit)					

Length: Header + data in bytes (minimum 8 for empty data). Checksum: Optional in IPv4, mandatory in IPv6.

Advantages over TCP: Lower overhead (8 bytes vs 20+), no connection setup, no ACK delay.
Limitations: No reliability, no ordering, no congestion control.

Q4) b) Sockets

[6]

Socket is an endpoint for communication between two machines. Identified by IP address + port number.

Types:

1. **Stream Socket (SOCK_STREAM):** TCP --- reliable, connection-oriented, ordered
2. **Datagram Socket (SOCK_DGRAM):** UDP --- unreliable, connectionless
3. **Raw Socket (SOCK_RAW):** Direct IP access (for custom protocols, ping)

Socket Functions (Connection-Oriented --- TCP):

```
sequenceDiagram
    participant Server
    participant Client

    Server->>Server: socket() --- create endpoint
    Server->>Server: bind() --- assign address & port
    Server->>Server: listen() --- wait for connections
    Client->>Client: socket() --- create endpoint
    Client->>Server: connect() --- request connection
    Server->>Client: accept() --- accept connection
    Note over Server,Client: TCP 3-way handshake
    Client->>Server: send() / write()
    Server->>Client: recv() / read()
    Server->>Client: close() --- terminate
    Client->>Client: close() --- terminate
```

Q4) c) SCTP Protocol

[6]

SCTP (Stream Control Transmission Protocol) is a reliable, message-oriented transport protocol.

Features:

1. **Multihoming:** An SCTP endpoint can have multiple IP addresses for redundancy
2. **Multistreaming:** Multiple independent streams within one connection (avoids head-of-line blocking)
3. **Message-oriented:** Preserves message boundaries (unlike TCP's byte stream)
4. **Four-way handshake:** Uses COOKIE mechanism to prevent SYN flood attacks
5. **Selective ACK:** Only missing chunks are retransmitted

Comparison: | Feature | TCP | UDP | SCTP | |-----|-----|-----| | Reliability | Yes | No | Yes | | Ordered delivery | Yes | No | Optional | | Multistreaming | No | No | Yes | | Multihoming | No | No | Yes |

Q5) a) HTTP

[9]

HTTP (HyperText Transfer Protocol) is an application-layer protocol for distributed, collaborative hypermedia systems (the web).

HTTP Request Message Format:

```
METHOD /path HTTP/version\r\n
Header1: value\r\n
Header2: value\r\n
\r\n
body (optional)
```

HTTP Response Message Format:

HTTP/version StatusCode StatusText\r\n
Header1: value\r\n
\r\n
body (HTML/content)

HTTP/1.1 vs HTTP/2:

Feature	HTTP/1.1	HTTP/2
Multiplexing	No (one request per connection)	Yes (multiple streams over one connection)
Header compression	No (plain text headers)	HPACK compression
Server push	No	Server can push resources proactively
Binary protocol	Text-based	Binary
Latency	Head-of-line blocking	Reduced latency via multiplexing

Q5) b) SMTP and MIME [9]

SMTP (Simple Mail Transfer Protocol): Used for sending emails between mail servers.

- Port: 25 (default), 587 (submission)
- Commands: HELO, MAIL FROM, RCPT TO, DATA, QUIT
- Push protocol (sender pushes to receiver)

MIME (Multipurpose Internet Mail Extensions): Extends SMTP to support non-text data.

- Types: text/plain, text/html, image/jpeg, application/pdf
- Encoding: Base64, quoted-printable

Email Delivery Flow:

Sender → MUA (Outlook) → MSA (port 587) → MTA (SMTP) →
→ MDA → POP3/IMAP server → MUA (Receiver)

POP3 (Post Office Protocol v3): Downloads emails to the client, deletes from server.

Webmail: Browser-based email access via HTTP (uses IMAP for server-side storage).

Q6) a) DHCP [9]

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration to hosts.

Client State Diagram:

```
stateDiagram-v2
    [*] --> INIT : Client boots
    INIT --> SELECTING : DHCPDISCOVER broadcast
    SELECTING --> REQUESTING : DHCPOFFER received
    REQUESTING --> BOUND : DHCPACK received
    BOUND --> RENEWING : 50% lease time passed
    RENEWING --> BOUND : DHCPACK (unicast)
    RENEWING --> REBINDING : 87.5% lease time passed
    REBINDING --> BOUND : DHCPACK (broadcast)
    REBINDING --> INIT : Lease expired / DHCPNACK
    BOUND --> INIT : Release
```

DORA Process:

1. **Discover** --- Client broadcasts "I need an IP"
2. **Offer** --- DHCP server offers an IP
3. **Request** --- Client requests the offered IP
4. **Ack** --- Server acknowledges

Q6) b) DNS and FTP

[9]

i) **DNS (Domain Name System):** Resolves human-readable domain names to IP addresses.

Recursive Query: DNS server resolves the query completely (may query other servers on behalf of client).

Client → Local DNS → Root DNS → TLD DNS → Authoritative DNS → IP

Iterative Query: DNS server returns the best answer or referral to the next server.

Client → Local DNS → (referral to root) → Client queries root → (referral to .com) → ...

ii) **FTP (File Transfer Protocol):** Transfers files between client and server.

- **Control connection:** Port 21 (persistent, carries commands/responses)
- **Data connection:** Port 20 (for actual file transfer, created per transfer)
- **Active mode:** Server connects to client's data port
- **Passive mode:** Client connects to server's data port (for clients behind NAT)

Q7) a) X.800 Security Architecture

[6]

ITU-T X.800 defines a comprehensive framework for network security.

Three main concepts:

1. Security Attacks:

- **Passive:** Eavesdropping, traffic analysis
- **Active:** Masquerade, replay, modification, denial of service

2. Security Services:

- **Authentication:** Verifies identity (peer entity, data origin)
- **Access Control:** Prevents unauthorized use
- **Data Confidentiality:** Protects from unauthorized disclosure
- **Data Integrity:** Ensures data is not modified
- **Non-repudiation:** Sender/receiver cannot deny involvement

3. Security Mechanisms:

- Encipherment, digital signatures, access control, data integrity
- Authentication exchange, traffic padding, routing control, notarization

X.800 Architecture		
Attacks	Services	Mechanisms
Passive	Authentication	Encipherment
Active	Access Control	Digital Signatures
	Confidentiality	Access Control Lists
	Integrity	Authentication Exchange
	Non-repudiation	Traffic Padding

Q7) b) HTTPS

[6]

HTTPS (HTTP Secure) is **HTTP over TLS/SSL** --- encrypts all communication between browser and web server.

- **Port:** 443
- **Protocol:** HTTP + SSL/TLS
- **Encryption:** Symmetric (session key) + Asymmetric (certificate exchange)
- **Certificate:** X.509 digital certificate binds identity to public key

Benefits:

- Confidentiality (no eavesdropping)

- Integrity (no tampering)
- Authentication (verifies server identity via CA-signed certificates)

Q7) c) Intrusion Detection Systems (IDS)

[5]

IDS monitors network traffic and system activity for malicious behavior.

Types:

1. **NIDS (Network-based)**: Monitors network traffic passively
2. **HIDS (Host-based)**: Monitors OS logs, file integrity on a single host
3. **DIDS (Distributed)**: Combines NIDS + HIDS sensors with central analysis

Detection Methods:

- **Signature-based**: Matches known attack patterns (low false positives, misses zero-day)
- **Anomaly-based**: Detects deviations from normal behavior (catches zero-day, high false positives)
- **Hybrid**: Combines both approaches

Q8) a) Symmetric vs Asymmetric Cryptography

[6]

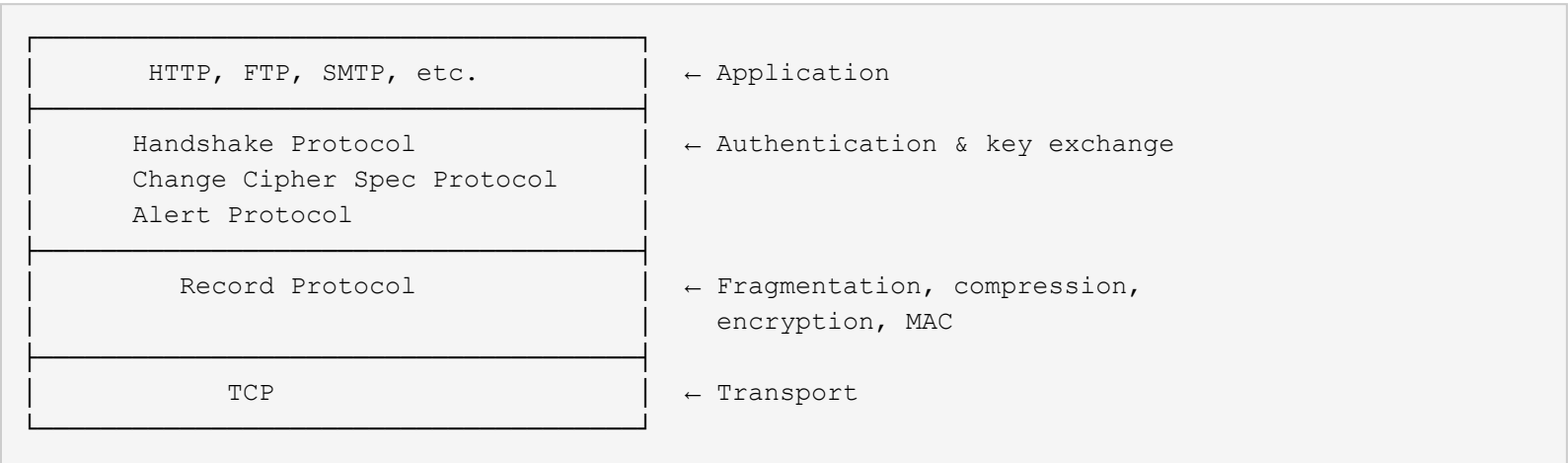
Parameter	Symmetric	Asymmetric
Keys	Single shared key	Public/private key pair
Speed	Fast (suitable for bulk data)	Slow (100-1000× slower)
Key exchange	Problematic (must share securely)	No shared secret needed
Key size	128-256 bits	2048-4096 bits
Security service	Confidentiality	Confidentiality + authentication
Algorithms	AES, DES, 3DES, Blowfish	RSA, ECC, Diffie-Hellman
Use case	Data encryption at rest	Key exchange, digital signatures

Q8) b) SSL (Secure Socket Layer)

[6]

SSL provides secure communication between client and server.

SSL Protocol Stack:



SSL Handshake (simplified):

1. Client sends **ClientHello** (SSL version, cipher suites, random)
2. Server responds **ServerHello** (chosen cipher, certificate, random)
3. Client verifies certificate, generates **pre-master secret**, encrypts with server's public key
4. Both compute **master secret** and **session keys**
5. Client sends **ChangeCipherSpec** + **Finished** (encrypted)
6. Server sends **ChangeCipherSpec** + **Finished** (encrypted)
7. **Secure channel established**

TLS is the successor to SSL (TLS 1.3 is current standard).

Firewall is a network security system that monitors and controls incoming/outgoing traffic based on predetermined rules.

Types:

1. **Packet Filter Firewall:** Examines packet headers (IP, port, protocol). Fast but stateless.
2. **Stateful Inspection Firewall:** Tracks connection state. More secure than packet filter.
3. **Application Gateway (Proxy):** Inspects application-layer data. Most secure but slowest.
4. **Next-Gen Firewall (NGFW):** Combines firewall + IDS/IPS + application awareness.

Architectures:

- **Screened host:** Single firewall protects the entire network
- **DMZ (Demilitarized Zone):** Three-leg firewall --- public servers in DMZ, internal network protected

Advantages: Traffic filtering, network segmentation, logging, attack prevention.

EXAMINER COMMENTARY

Why this scores full marks:

- Subnetting answer includes complete hex-to-decimal conversion steps
- UDP dump parsed field-by-field from hex to decimal
- Tables used for all comparisons (circuit vs packet, TCP vs UDP vs SCTP, symmetric vs asymmetric)
- Diagrams for TCP/UDP headers, SSL stack, firewall architectures
- X.800 answer shows the three-part architecture clearly
- Socket programming includes sequence diagram

Common Deductions:

- Forgetting to show the binary conversion in subnetting problems
- Mistaking UDP header length for data length
- Confusing HTTP request vs response format
- Mixing up POP3 vs IMAP behavior (POP3 downloads + deletes; IMAP manages server-side)
- Not distinguishing between symmetric key exchange problem and asymmetric solution
- Omitting the checksum in TCP/UDP header diagrams

Time Budget:

- Q1 (18 min): Circuit/Packet 6 min + RIP 6 min + Subnetting 6 min
- Q2 (18 min): IPv6 6 min + BGP 6 min + Network layer 6 min
- Q3 (18 min): TCP header 6 min + Transport services 6 min + UDP parsing 6 min
- Q4 (18 min): UDP header 6 min + Sockets 6 min + SCTP 6 min
- Q5 (18 min): HTTP 9 min + SMTP/MIME 9 min
- Q6 (18 min): DHCP 9 min + DNS/FTP 9 min
- Q7 (17 min): X.800 6 min + HTTPS 6 min + IDS 5 min
- Q8 (17 min): Crypto 6 min + SSL 6 min + Firewall 5 min
- **Total: ~142 min** (within 150 min limit)